First Response:

A) ANCHOR SCENES

SCENE #1: First Alzheimer's Mutation Discovered (Amyloid Cascade Hypothesis)

- **Summary:** In the late 1980s, Hardy investigated an early-onset Alzheimer's family (led by Carol Jennings) that had multiple members developing dementia in their 50s. By 1990, after years of genetic sleuthing, his team identified a mutation in the amyloid precursor protein (APP) gene in this family – the first genetic cause of Alzheimer's disease. This finding led Hardy (with colleague David Allsop) to propose the "amyloid cascade hypothesis" in 1992, positing that amyloid protein accumulation is the trigger for Alzheimer's pathology.

- **Date/Place: 1985–1991**, St. Mary's Hospital, London (Imperial College); findings published 1991–1992.

- **People:** John Hardy; Carol Jennings (family member who sought answers); research team including Alison Goate (postdoc who led gene sequencing); collaborator David Allsop.

- **Concrete details:** After Carol reached out in the mid-1980s about her stricken family, Hardy's lab collected their blood samples. They performed linkage analysis (focusing on chromosome 21 due to hints from Down syndrome and other studies) and painstaking DNA sequencing. In February 1991, they found a single missense mutation in the APP gene (codon 717) segregating with the disease. The team published this landmark discovery in *Nature* in 1991 – the so-called "London mutation" of APP. Hardy recalled the irony that a prior false lead from Massachusetts General Hospital had pointed to chromosome 21; it turned out "wrong in the right place," as his finding confirmed APP's role. In 1992, Hardy and colleagues formally proposed that amyloid-beta accumulation triggers a cascade including tau tangles and neuron death.

- **Why it matters:** This was the first evidence linking a specific gene mutation to Alzheimer's disease, transforming the field. It cemented the idea that a biochemical change – excessive amyloid-beta peptide from mutant APP – could drive neurodegeneration. Hardy's amyloid cascade hypothesis provided a unifying framework for Alzheimer's pathology (plaques, tangles, cell death), guiding decades of research and drug development attempts. Career-wise, this discovery made Hardy a leader in neurogenetics and earned him major awards (e.g. the 1993 Potamkin Prize).

- **Evidence quality: Primary and strong.** Multiple first-hand sources document this scene: Hardy and team's 1991 *Nature* paper (most cited of that year), Hardy's own recollections (interviews), and institutional histories. UCL interviews quote Hardy on Carol's family and the 5-year effort to find the mutation. The amyloid hypothesis publication is directly credited to Hardy et al. 1992.

- **Myth risk: Low.** While often recounted, the story is backed by contemporary publications and Hardy's interviews, not anecdotal lore. (One should note Alison Goate was first author of the 1991 paper, sometimes underplayed in popular retellings, but the credit is well documented.) No "too-perfect" elements – the timeline and family involvement are confirmed by reliable sources.

- **Sources:** UCL News (2022); *Nature* (Goate et al. 1991) via analysis; Journal of Law & Biosci. (Bubela et al. 2015); Guardian (2018).

SCENE #2: Conflict Over the "Swedish Mutation" Patent

- **Summary:** In 1992, shortly after the London APP mutation, Hardy's team (now partly in Florida) discovered a second Alzheimer's mutation – a double mutation in APP in two Swedish families. However, a major falling-out occurred when collaborator Dr. Michael Mullan secretly patented this "Swedish mutation" without listing Hardy as co-inventor. The saga involved covert sequencing, a for-profit entity (Alzheimer's Institute of America, AIA) holding the patent, and years of legal battles. Hardy felt betrayed and eventually testified against the patent, which was invalidated for improper inventorship.

- **Date/Place: 1992–1993**, spanning Imperial College London and University of South Florida (Tampa). Key events: February 1992 in London (samples from Sweden analyzed); mid-1992 in Tampa (mutation sequenced and patented); fallout by 1993.

- **People:** John Hardy; Michael Mullan (clinical research fellow in Hardy's lab); Dr. Lars Lannfelt (Swedish collaborator who provided family samples); Fiona Crawford (team member); Ronald Sexton (venture capitalist backing the patent).

- **Concrete details:** After the London mutation, Dr. Lannfelt brought Hardy pedigree data and samples from two Alzheimer's-stricken Swedish families in early 1992. Hardy's student Henry Houlden ran a genetic linkage test, which indicated an APP gene defect in both families. Hardy then sent the DNA samples to Mullan in Florida for sequencing. Unbeknownst to Hardy, Mullan had Fiona Crawford perform the sequencing off-campus at a private lab (TBRI) to keep it outside university oversight. They identified two adjacent mutations (the "Swedish APP670/671" mutation) that dramatically increase

amyloid-beta production. Mullan rushed to publish the finding in *Nature Genetics* in 1992, pointedly leaving Hardy off the author list (by prior agreement between them). Only Swedish funding agencies were acknowledged, omitting Hardy's UK funding. At virtually the same time (May 1992), Mullan and Sexton incorporated AIA and filed patents claiming the mutation, naming Mullan as sole inventor. Hardy, still in London, was kept in the dark; an arranged letter to waive University of South Florida's rights obscured the invention's timing. In 1993, Hardy accidentally discovered correspondence about commercially licensing a mouse model with the Swedish mutation – realizing Mullan's covert patent move. Feeling cheated, Hardy cut ties with Mullan; their relationship "became strained" amid accusations. Years later, in patent litigation, a judge concluded that leaving Hardy's name off was deliberate "to avoid the inventions becoming the property of Imperial". Hardy served as a key witness against AIA, and courts ultimately invalidated Mullan's patent for failing to include Hardy and others as co-inventors.

- **Why it matters:** This controversy highlighted issues of ethics and ownership in science. The Swedish mutation became crucial in Alzheimer's research (used in transgenic mouse models globally), yet its patent was used by AIA as a tool to sue researchers for royalties. The dispute likely altered Hardy's career course: disillusioned with Imperial and the patent wrangling, he and his team moved fully to the U.S. (Mayo Clinic by 1996). It also exemplifies how collaboration can sour over credit and profit – a cautionary tale for the research community. Scientifically, the Swedish mutation's discovery (despite the conflict) reinforced the amyloid hypothesis by providing a second familial AD mutation that heightened amyloid production.

- **Evidence quality: Primary/legal and strong.** The sequence of events is documented through legal records and detailed case studies. A 2015 Journal of Law & Biosciences analysis (Bubela *et al.*) reconstructs the timeline from court findings and interviews. Hardy and Mullan's actions are recorded in trial testimony and contemporaneous correspondence (e.g. the patent filing, the Nature Genetics paper). Hardy's own comments appear in sources like *Decoding Darkness* (via Tanzi) and legal depositions. The patent was invalidated in court (Eastern District of PA, 2012) for improper inventorship, a fact reported in legal summaries. This is not hearsay but derived from judicial decisions and first-person accounts.

- **Myth risk: Low-medium.** The narrative is complex, and simplified versions sometimes circulate without full context. For example, Mullan often claimed Hardy "didn't believe the Swedish family had a mutation" as justification for patenting it alone – a one-sided claim from an interested party. We flag this because it's repeated in Tanzi's *Decoding Darkness* but is not corroborated by Hardy (who clearly invested in analyzing those

samples). The "secret patent" story might sound sensational, but it's well supported by the evidence (court documents and emails). It's important to rely on the documented facts (e.g. Hardy's omission as author/inventor was deliberate to avoid institutional claims) rather than any single participant's convenient narrative.

- **Sources:** Bubela *et al.*, *J. Law & Biosci.* (2015); Bubela *et al.* (court findings); Hardy interview via Tanzi (2000); Guardian (2018); U.S. District Court decision summarized in RealClearInvestigations (2018).

SCENE #3: New Genes, New Diseases – Tau and Parkinson's Discoveries

- **Summary:** Building on his Alzheimer's work, Hardy went on to help crack other neurodegenerative diseases in the late 1990s. In 1998, while at Mayo Clinic Jacksonville, he was part of the international consortium that identified mutations in the tau protein gene (MAPT) as a cause of frontotemporal dementia (Pick's disease). A few years later, at NIH, Hardy's team co-discovered that an *extra* copy of the alpha-synuclein gene (a rare "triplication" mutation) causes an inherited form of Parkinson's disease. These findings broadened the "protein misfolding" paradigm from Alzheimer's to other dementias.

- **Date/Place: 1998**, Mayo Clinic (Jacksonville, FL) – tau mutation in familial FTD; **2003**, NIH/National Institute on Aging (Bethesda, MD) – alpha-synuclein triplication in Parkinson's (published 2003).

- **People:** John Hardy (as consortium leader/member); Mike Hutton and others (co-discoverers of tau mutation); Dr. Peter Heutink, Dr. Michel Goedert (other tau researchers); Andrew Singleton and Dr. Mihael Polymeropoulos (co-discoverers of alpha-synuclein mutations in Parkinson's).

- **Concrete details:** By the late '90s, scientists suspected that tau protein (which forms tangles inside neurons) might directly cause neurodegeneration. Hardy joined forces with groups in the UK and Netherlands to hunt for tau gene mutations in families with hereditary frontotemporal dementia. In 1998, they found missense and splice-site mutations in MAPT (tau) in several families, proving tau abnormalities alone could drive dementia. The landmark paper (Hutton *et al.*, *Nature* 1998) had over 20 co-authors including Hardy. Hardy's genetic approach thus extended beyond Alzheimer's plaques to the tangles of FTD. Next, after moving to NIH in 2001, Hardy collaborated on Parkinson's genetics. In 2003, his NIH team (including Andrew Singleton) discovered that some Parkinson's patients have a duplication or triplication of the alpha-synuclein gene, leading to toxic accumulation of the synuclein protein. This was evidenced by a

large family with multiple affected members—those who inherited three copies of the gene developed early Parkinson's symptoms. It was a striking finding because it wasn't a point mutation but a gene dosage effect causing a protein aggregation (Lewy bodies) disease. Hardy's lab confirmed that having an extra *normal* gene could cause pathology, reinforcing the idea that overproduction of a misfolding-prone protein can kill neurons (analogous to APP in Alzheimer's).

- **Why it matters:** These discoveries showed that Hardy's "genetic strategy" could elucidate multiple neurodegenerative diseases. Finding *tau* mutations was crucial because it validated that tangles (tau) are not just byproducts but can be primary drivers of dementia. In Alzheimer's research, this spurred a "tau vs amyloid" debate – Hardy's own work ironically provided a key to the alternative hypothesis (tau pathology), although he remained an amyloid champion. The Parkinson's gene finding linked another misfolded protein (alpha-synuclein) to disease, cementing the concept that different protein aggregates cause different neurodegenerative disorders. Hardy's role in these breakthroughs earned him recognition beyond Alzheimer's – he received awards like the 2008 Anne Oprecht Prize for Parkinson's research. These scenes also demonstrate Hardy's leadership in team science: large collaborative genetics projects that mapped disease mechanisms, a hallmark of his career.

- **Evidence quality: Secondary with primary references, strong.** Hardy's involvement is confirmed by authoritative sources: the Cure Alzheimer's Fund biography (which he authored or approved) explicitly notes "he was part of the consortium which identified mutations in the tau gene in Pick's disease (1998)" and "part of the group which found triplications in the synuclein gene causing Parkinson's". The original papers in *Nature* (1998) and *Science* (2003) list him as co-author. His Wikipedia entry and award citations also reference these findings. While the narrative detail is sparser than for the Alzheimer's scenes, the core facts are well documented in the scientific literature.

- **Myth risk: Low.** These are straightforward scientific milestones with clear documentation. There is little mythologizing around Hardy's role – he himself often credits the "consortium" or team. One subtle point: some popular accounts focus solely on Hardy's amyloid work and omit these contributions; conversely, these genetics breakthroughs are sometimes reported without mentioning Hardy (since they involved many players). In a narrative, one should avoid overstating Hardy as a lone discoverer – he was a key part of larger efforts. No dubious anecdotes are attached; the importance of tau and synuclein was recognized with Hardy sharing prizes like the Brain Prize 2018 in part for this body of work.

- **Sources:** Cure Alzheimer's Fund profile; Nature (1998) via Wikipedia; *Science* (Singleton et al. 2003) via profile; Hardy – Awards list (Royal Society).

SCENE #4: 30-Year Vindication – First Drug Targeting Amyloid Shows Benefit

- **Summary:** In November 2022, results from a clinical trial of the drug lecanemab marked the first time an anti-amyloid therapy significantly slowed Alzheimer's disease progression. For John Hardy, this news was a triumphant validation of the hypothesis he proposed three decades earlier. At age 68, Hardy watched as the "modest" 27% slowing of decline was hailed as *historic*, proving that clearing amyloid plaques can benefit patients. Hardy called it "the beginning of the end" for Alzheimer's – an emotional culmination of his life's work.

- **Date/Place: November 29, 2022**, Clinical Trials on Alzheimer's Disease (CTAD) conference in San Francisco (where results were presented) and London (Hardy's commentary at UCL after publication in *NEJM*).

- **People:** John Hardy; the lecanemab trial investigators (Eisai/Biogen); UCL colleagues like Nick Fox and Cath Mummery; Alzheimer's patients and families awaiting breakthroughs.

- **Concrete details:** The lecanemab Phase 3 trial tested an antibody that removes beta-amyloid plaques. In late 2022, researchers announced that over 18 months, lecanemab slowed cognitive decline by ~27% in early Alzheimer's patients (1,795 participants). Hardy, as the originator of the amyloid theory, was asked to comment. He described the outcome as both "modest and historic" – modest in effect size, but historic as a proof of concept. In a UCL press release, Hardy said: "This trial is an important first step, and I truly believe it represents the beginning of the end…It's fantastic to receive this confirmation that we've been on the right track all along". He noted it took 30 years to reach this point, reflecting on how his Alzheimer's Society-funded research in 1989 first implicated amyloid. Hardy did temper the excitement with caution about side effects (some patients had brain swelling or bleeds), urging rigorous monitoring as this new class of drugs is rolled out. Still, the day the results came out, he reportedly celebrated the validation of the amyloid hypothesis, feeling that a turning point in the disease had finally arrived. Shortly thereafter, regulators began reviewing lecanemab, and the Alzheimer's research community felt a renewed optimism – much of which Hardy, as a newly knighted Sir John, lived to see.

- **Why it matters:** This scene provides narrative closure – the idea Hardy championed (often amid skepticism) was proven correct in the clinic. It underscores the theme of

perseverance in science: Hardy's initial discovery in 1991 led to a hypothesis, years of trial failures followed, but ultimately a therapy vindicated the approach. For a biochemistry audience, it connects molecular discovery to real-world impact. Human-wise, Hardy's satisfaction is palpable; he specifically credited the "selfless dedication of families" who participated in research, including those like Carol Jennings's family who contributed to the 1989 discovery. This moment also signals a new chapter – as Hardy said, it's just the "first step," implying there's more work to do but with proof that the strategy can work. In a narrative, this could serve as a powerful concluding scene, showing the payoff of decades of scientific toil and tying back to the opening scene's discovery.

- **Evidence quality: Primary (news/interview) and strong.** Hardy's statements are taken directly from a UCL News feature that quoted him on the trial day. The trial data itself was published in *NEJM* (Nov 2022) and is not in dispute. Multiple media outlets reported Hardy's reaction; the UCL piece and a Guardian article capture his quotes and emotional stance. As a first-person reaction by Hardy to a public scientific result, the sourcing is reliable.

- **Myth risk: Low.** The events are very recent and documented in real time. There is little opportunity for myth-making yet; Hardy's "beginning of the end" line is dramatic, but it's a direct quote. One should avoid overstating what lecanemab does (it's not a cure, only a slowing), something Hardy himself is careful about. As long as the scene is presented with that nuance (a milestone, not a miracle), it remains factual. This scene is a nice counterpoint to any earlier doubts – it's grounded in peer-reviewed trial evidence, not an anecdote.

- **Sources:** UCL News (Hardy on lecanemab, 2022); *NEJM* 2023 (lecanemab study) via UCL summary; Guardian (Devlin, 2018) – context of past trial failures.

B) DIRECT QUOTES

1. **"Throughout her family were people who developed Alzheimer's and did so in their 50s, and she wanted answers. She's a very determined woman, a remarkable woman."**
   Context/Date: Recalling how patient advocate Carol Jennings first contacted him in the mid-1980s about her family's early-onset Alzheimer's; quoted in a UCL News interview, December 2022. Hardy is describing the motivation and tenacity of the family that

sparked his APP mutation research.
Source: UCL News feature (John Hardy interview), 2 Dec 2022.

2. **"Bizarre, because it was wrong in the right place."**
Context/Date: Hardy reflecting on the peculiar fact that an early linkage study (which erroneously linked Alzheimer's to chromosome 21) still pointed him to the correct gene (APP) in hindsight. Quoted in the book *Decoding Darkness* (c. 2000), recounting his thoughts on the 1991 APP mutation discovery. This line conveys Hardy's wry realization that a false lead led them to the right answer.
Source: T. Bubela *et al.*, *J. Law and Biosciences* 2(2):213-262 (2015), quoting Hardy's recollection.

3. **"We haven't got treatments and therapies, let's be honest… We haven't scored any goals and so talking about who has been the best goal scorer is a bit premature really."**
Context/Date: Hardy in a December 2015 interview (after winning the Breakthrough Prize), cautioning against self-congratulation in the absence of an actual cure. He uses a football metaphor to emphasize that despite genetic discoveries, no effective therapies existed yet, urging modesty among researchers.
Source: Neurology Central interview (Alice Weatherston) with John Hardy, 8 Dec 2015.

4. **"The first conference only attracted 40 people, but now thousands attend."**
Context/Date: Comment made by Hardy at a press conference on 6 March 2018 (upon receiving the Brain Prize). He is contrasting the state of Alzheimer's research in the early 1980s – when it was a scientific backwater – to the huge field it has become by the 2010s. It underscores how far the research community has grown since he started.
Source: The Guardian (Hannah Devlin), "Brain prize winner calls Brexit a 'disaster'…" 6 Mar 2018.

5. **"It's like popping a statin to treat a heart attack – the right drug but the wrong time."**
Context/Date: Hardy's analogy in the same March 2018 discussion, addressing why amyloid-clearing drugs had failed in past trials. He argued the concept was right but they were given too late in the disease course. This quote reveals Hardy's way of explaining scientific issues in simple, relatable terms (comparing late-stage Alzheimer's treatment to taking cholesterol medicine during a heart attack).
Source: The Guardian (H. Devlin) report on Brain Prize press event, 6 Mar 2018.

6. **"This trial is an important first step, and I truly believe it represents the beginning of the end. The amyloid theory has been around for 30 years so this has been a long**

time coming."
Context/Date: Hardy's reaction to the positive lecanemab drug trial, quoted in a UCL press release on 2 Dec 2022. He speaks as someone who formulated the amyloid hypothesis 30 years prior, now seeing validation. This quote captures his hope and sense of vindication at this turning point in Alzheimer's treatment.
Source: UCL News – John Hardy on lecanemab trial results, 2 Dec 2022.

C) HUMAN REALITY DETAILS

- **Early spark:** Hardy's interest in biochemistry was ignited by a high school teacher (Mrs. Cox) in the late 1960s, who encouraged his fascination with biology. This mentorship set him on the path to a science career. — *Source: Dementia Researcher profile (NIHR), 2023.*

- **Biochemist by training:** He earned a B.Sc. in Biochemistry from the University of Leeds in 1976 and a Ph.D. in Neuropharmacology from Imperial College London in 1981, where he studied dopamine and amino acids in the brain. This strong biochemistry foundation later helped him probe the molecular basis of Alzheimer's. — *Source: NIHR profile, 2023.*

- **Postdoc in a prion lab:** In the early 1980s, Hardy did postdoctoral research at the MRC Neuropathogenesis Unit (Newcastle, UK), which was renowned for work on scrapie prions, and then at the Swedish Brain Bank in Umeå. In Sweden he began focusing on Alzheimer's disease, getting hands-on experience with brain tissue and familial cases that shaped his genetic approach. — *Source: NIHR profile, 2023.*

- **Alzheimer's was a backwater:** When Hardy entered the field, Alzheimer's research was tiny. "The first conference only attracted 40 people," he noted. By contrast, today's AD conferences draw thousands – a growth he witnessed and helped bring about through his high-impact discoveries. — *Source: Guardian interview, 2018.*

- **Global career moves:** Hardy's career spanned continents. He started in London academia, then moved to the US in 1992 (University of South Florida), later leading labs at Mayo Clinic and the NIH, and finally returned to University College London in 2007. These moves often followed opportunities (or frustrations) in funding and infrastructure, illustrating the challenges of sustaining big-genetics research. — *Source: NIHR profile*

timeline.

- **Family-inspired science:** Hardy's pivotal Alzheimer's mutation discovery began with a direct outreach from a family seeking answers. He formed a close collaboration with Carol Jennings's family, whose willingness to contribute blood samples over years made the breakthrough possible. This underscores how patient families influenced his research trajectory in a very personal way. — *Source: UCL News (Hardy interview), 2022.*

- **Outspoken advocate:** Hardy is unafraid to voice political views when science is at stake. Upon winning the Brain Prize in 2018, he called Brexit "an unmitigated disaster for science and the health service" and pledged to donate part of his €1M prize to an anti-Brexit group. He often speaks up about funding and policy issues that affect research progress. — *Source: The Guardian, Mar 2018.*

- **Principled critic:** He has openly criticized pharmaceutical companies for abandoning Alzheimer's research after drug failures. In early 2018, as Pfizer pulled out of neuroscience, Hardy lamented the "great disappointment" of trial failures but argued "pharma is making a mistake" by quitting – insisting that the amyloid approach was still right if applied earlier. This candor reflects his determination and leadership in the field, pushing others not to lose hope. — *Source: The Guardian, 2018.*

- **Mentorship and lab culture:** Many of Hardy's trainees have become prominent scientists. Notably, his first postdoc hire, Alison Goate, was lead author on the 1991 APP paper and is now a leading Alzheimer's geneticist herself. Colleagues describe Hardy as collaborative and generous with data, forming international teams rather than hoarding credit. The race to find AD genes in the '80s and '90s, in which Hardy participated, has been described as "a mixture of idealism, selfishness, generosity, greed, fun, anger, sex, drugs and rock 'n' roll" – reflecting the intense, colorful environment he navigated. — *Source: J. Law & Biosci. (2015); Nature Medicine profile (2004).*

- **Personal life:** Hardy has three adult children and three grandchildren, all living in the United States. Balancing transatlantic family ties, he spent a significant portion of his career in American institutions before returning to London. (His familial status is rarely mentioned in scientific contexts, but it humanizes the man behind the science.) — *Source: Cure Alzheimer's Fund profile, updated 2019.*

- **Street-food aficionado:** A fun personal fact – Hardy enjoys trying street food wherever he travels. This quirky detail (shared in a 2023 profile Q&A) shows a down-to-earth side; despite knighthood and awards, he's game for a humble food stall in any country. —

Source: NIHR "Meet the Researchers" profile, 2023.

- **Media-shy about social media:** Unlike many modern scientists, Hardy maintains a low profile online – he doesn't use Twitter or Instagram for science outreach. Instead, he lets his research speak for itself in publications and interviews. — *Source: NIHR profile Q&A, 2023.*

D) SCIENCE CONNECTION (Intro Biochem Bridge)

- **Best-fit topic: Protein structure & aggregation in metabolism (proteolysis and neurodegenerative disease)** – Hardy's work connects DNA mutations to protein misfolding and aggregation (amyloid plaques and tau tangles), a theme relevant to protein chemistry and enzymatic processing of proteins.

- **Precise contribution:** *John Hardy discovered the first gene mutation (in APP) that causes early-onset Alzheimer's, demonstrating that an abnormal cleavage product of a protein (amyloid-β peptide) can accumulate and trigger a cascade of neurodegenerative processes.*

- **Bridge explanation:** In biochemical terms, Hardy's discovery showed how a change at the DNA level can alter protein metabolism and lead to disease. The APP gene encodes the amyloid precursor protein, which is normally cut by enzymes (secretases) to produce a small peptide, Aβ. Hardy's identified mutation makes this cut occur in a way that produces excess sticky Aβ. The surplus peptide clumps into insoluble plaques between neurons. These amyloid deposits then set off a chain reaction: they overstimulate neurons and immune cells (microglia), causing chronic inflammation. They also somehow induce tau protein to misfold into tangles inside neurons, leading to cell death. Intro biochemistry students can see multiple concepts here – proteolytic processing of a membrane protein, protein aggregation due to altered structure, and the downstream effects on cellular metabolism (glial activation and oxidative stress). Hardy's amyloid cascade hypothesis elegantly ties a single molecular event (peptide accumulation) to complex pathological outcomes (brain atrophy and dementia).

- **Scientist's own framing/analogy:** Hardy often uses analogies to explain these ideas. He described the Alzheimer's pathology as a "car crash" before his work helped "put that

pathology into order" – meaning they figured out what comes first. By identifying the APP mutation, Hardy could say: instead of seeing a jumble of plaques and tangles, we now know the "amyloid hit" happens first, driving the rest. He also likened treating late-stage Alzheimer's to "popping a statin during a heart attack" – the right mechanism (lowering cholesterol/amyloid) but too late to prevent damage. These analogies help students grasp the importance of timing and sequence in biochemical disease processes.

- **Sources:** UCL News (mechanism of APP mutation); UCL News (amyloid cascade description); Guardian 2018 (Hardy analogies); Cure Alzheimer's Fund bio (Hardy's contribution).

E) COMPLICATIONS OR CONTROVERSIES

- **Item:** *Patent Dispute over APP "Swedish Mutation" (Hardy vs. Mullan/AIA)*.
  What's agreed upon: Hardy's group discovered the "Swedish" double mutation in APP in 1992 – a finding published with Mullan and others, but notably without Hardy as co-author. Shortly after, Dr. Michael Mullan filed patents on this mutation, assigning them to the Alzheimer's Institute of America (AIA), and did not list Hardy or the Imperial College team members as inventors. AIA then enforced these patents aggressively, demanding licensing fees from researchers using the mutation in Alzheimer's mouse models. It's agreed that Hardy and Mullan had a falling-out by 1993 over this issue, and Hardy ended up opposing the patents in court. Legal proceedings in the 2000s revealed that the patent filings excluded co-inventors and that the work had been done partly under Imperial College's purview. Ultimately, U.S. courts invalidated the AIA patents in 2012 for incorrect inventorship (ruling that Hardy *et al.* should have been included).
  What's disputed: The main disputes were (1) credit and inventorship – Mullan claimed sole credit, whereas Hardy's side argued it was a team effort; and (2) institutional rights – whether Imperial College or USF had rights to the discovery, which Mullan tried to circumvent. Mullan has alleged that Hardy wasn't convinced the Swedish families even had a mutation and thus didn't initially prioritize them, implying Hardy's skepticism justified Mullan's actions. Hardy has not concurred with that characterization; he was actively involved up to the sequencing stage and was upset by being cut out. Another point of contention was whether USF (Mullan's employer in 1992) was aware of the invention; a letter was obtained to waive USF's claims, but its scope was later contested

in court. Essentially, the motives and honesty of Mullan's actions were disputed – in court, Hardy's side portrayed it as a deliberate "conspiracy to avoid…Imperial College or Athena (company) ownership", while Mullan insisted he was simply protecting what he discovered.

Best sources: *Journal of Law and Biosciences* (2015) by Bubela *et al.* gives a comprehensive, neutrally-toned case study. It includes court findings, which favored Hardy's perspective (e.g. Judge's conclusion that leaving Hardy off was intentional). Another source is the book *Decoding Darkness* (Tanzi & Parson, 2000), which presents insiders' accounts (Tanzi quotes Mullan and Hardy) – though one should be cautious as it reflects personal viewpoints. The U.S. District Court decision (Eastern District of PA, 2012) is public record; it voided the patent largely due to improper inventorship (Hardy being omitted). News outlets like *Nature* (Check Hayden, 2011) reported on the lawsuit's collapse, confirming the outcome. In summary, the best understanding comes from legal documents (for facts) and Hardy's own comments in interviews about how betrayed he felt. Both sides of the dispute (Hardy vs. Mullan) are represented in court transcripts and Tanzi's narrative, providing a balanced view if both are consulted.

F) "NOT ON WIKIPEDIA" FINDS

- **Hidden patent saga –** Hardy's role in the *"mouse that trolled"* patent case is rarely mentioned on Wikipedia. The dramatic story of how colleague Michael Mullan secretly patented their joint discovery (the Swedish APP mutation), forming a "patent troll" entity (AIA) that sued other researchers, is a striking episode. It's distinctive because it highlights ethical and legal conflicts in science, showing another dimension of Hardy's career (standing up in court for open science) that casual bios overlook. — *Source: Bubela et al. 2015, J. Law & Biosci..*

- **Brexit outcry and prize donation –** After winning the Brain Prize in 2018, Hardy publicly blasted Brexit as "an unmitigated disaster for science" and pledged to donate part of his €1 million prize to anti-Brexit advocacy. This bold political stance (and use of prize money for activism) is not in his Wikipedia entry, yet it's a revealing detail about his character and the value he places on international collaboration. — *Source: Guardian, Mar 6, 2018.*

- **"Sex, drugs and rock 'n' roll" in AD research –** A colorful quote describing the Alzheimer gene-hunting race as "a mixture of idealism, selfishness, generosity, greed, fun, anger, sex, drugs and rock 'n' roll" doesn't appear on Wikipedia. It comes from

researchers of that era and captures the wild-west atmosphere Hardy operated in during the 1980s–90s. This humanizes the science history and contrasts with the dry tone of typical summaries. — *Source: J. Law & Biosci. 2015 (Bubela et al.), citing contemporaries.*

- **Street-food enthusiast –** Wikipedia doesn't tell you that Sir John Hardy is a street-food lover. In a 2023 profile, he shared that he enjoys trying street food around the world. It's a lighthearted detail that makes the eminent scientist more relatable (imagining him sampling satay or tacos at a food stall). Such personal trivia is often omitted from formal bios but can enliven a narrative profile. — *Source: NIHR Dementia Researcher Q&A, 2023.*

- **Family ties in the US –** Hardy's Wikipedia page lists his career and honors but not his family. Notably, he has three children and three grandchildren living in the United States. This transatlantic family fact isn't widely published, yet it might have influenced his decisions (e.g. spending many years working in America before returning to London). It adds a personal dimension — he's not just a scientist but also a father and grandfather balancing life across continents. — *Source: Cure Alzheimer's Fund profile.*

- **Most-cited AD researcher –** It's not on Wikipedia, but as of mid-2010s Hardy had over 23,000 citations and was the most cited Alzheimer's researcher in the UK (5th worldwide). This metric, mentioned in a 2015 analysis, underscores his influence in the scientific community. It's a distinctive accolade that goes beyond awards, highlighting the impact of his publications (though such stats can date quickly). — *Source: Bubela et al. in J. Law & Biosci. 2015.*

G) SOURCES CONSULTED

1. UCL News – *"Historic Alzheimer's breakthrough 30 years in the making"* (Dec 2, 2022). – *Feature article interviewing Prof. John Hardy.* **Source:** University College London Press Release. *(Provides Hardy's recollections of discovering the APP mutation with Carol Jennings's family, the development of the amyloid cascade hypothesis, and his reaction to the 2022 lecanemab trial results.)*

2. Devlin, Hannah. "Brain prize winner calls Brexit a 'disaster' for the NHS and science." – *The Guardian*, 6 Mar 2018. *(Covers John Hardy's comments at a 2018 press event: includes his*

quotes on the state of dementia research in the 1980s vs now, his anti-Brexit stance, and reflections on trial failures and the amyloid hypothesis (e.g. "right drug, wrong time" analogy).)

3. *John Hardy – Cure Alzheimer's Fund Profile.* *(Web profile last updated c.2019.)* **Source:** CureAlz.org. *(Provides a timeline of Hardy's education and career moves, and succinct statements of key accomplishments: finding the first APP mutation in 1991, tau gene discovery in 1998, alpha-synuclein triplication in Parkinson's. Also mentions personal details like number of children.)*

4. Bubela, Tania; Vishnubhakat, Saurabh; Cook-Deegan, Robert. "The Mouse That Trolled: The long and tortuous history of a gene mutation patent that became an expensive impediment to Alzheimer's research." – *Journal of Law and the Biosciences* **2**(2): 213–262 (July 2015). DOI:10.1093/jlb/lsv011. *(In-depth case study of the APP "Swedish mutation" patent saga. Used here to detail the Hardy–Mullan conflict: covers the discovery of the London and Swedish mutations, the patent filings and legal battles (Judge's findings), and context about the competitiveness of AD genetics research. Primary source for controversy details.)*

5. *Interview: John Hardy (Breakthrough Prize winner).* Neurology Central, 8 Dec 2015. *(Q&A by Alice Weatherston with John Hardy after he received the Breakthrough Prize. Hardy discusses his career and views. Quoted for his humility about lack of therapies.* ***Note:*** *Content accessed via Technology Networks transcript.)*

6. Mackenzie, Ruairi. "BNA Interview Series: The Genetics of Neurodegeneration With Professor John Hardy." – Technology Networks (Neuroscience), 26 Sept 2019. *(Transcript of an interview at the 2019 BNA Festival of Neuroscience. Hardy explains the impact of genetics on neurodegenerative disease understanding. Source for Hardy's technical insights (e.g. distinction between inherited vs sporadic Alzheimer's, microglial genes like TREM2). Used for context, though direct quotes in final text were from other sources.)*

7. NIHR Dementia Researcher. "Profile – Professor Sir John Hardy, University College London." (Interview/Profile, Nov 3, 2023). *(Short online profile with a mini-interview. Gave details on Hardy's early influences (Mrs. Cox, the teacher), education and career timeline, and personal Q&A (fun fact about street food, not on social media).)*

8. Wikipedia – *"John Hardy (geneticist)"* (last modified 2023). *(Used as a starting point for career chronology and awards. Specific data (e.g. FRS year, list of positions) cross-checked with other sources. No uncited claims taken directly.)*

9. Hardy, John et al. "Alzheimer's disease: the amyloid cascade hypothesis." – *Science* **256**:184-185 (1992). *(The seminal paper articulating Hardy's hypothesis. Not quoted directly here, but contextualized via secondary sources.)*

10. Goate, Alison et al. "Segregation of a missense mutation in the amyloid precursor protein gene with familial Alzheimer's disease." – *Nature* **349**:704–706 (21 Feb 1991). *(The original APP London mutation discovery paper. Provided factual basis for Scene #1 via secondary recountings. No direct text excerpt in our narrative, but foundational.)*

11. Tanzi, Rudolph & Parson, Ann. *Decoding Darkness: The Search for the Genetic Causes of Alzheimer's Disease*. (Perseus, 2000). *(This book offers an inside story of AD gene hunts. We relied on references to it within Bubela's article for quotes from Mullan and Hardy. It's a key source for the personal dynamics in the controversy.)*

12. RealClearInvestigations. "Famed Alzheimer's Researcher John Hardy Is a Knight...but Not in NIH's Good Graces." (J.P. London, Feb 2018). *(Investigative piece which mentions the AIA patent litigation outcome and NIH tensions. Cited for the patent's invalidation due to missing co-inventor Hardy. Used carefully due to potential bias; cross-referenced with academic source #4.)*

(Additional references for verification: Breakthrough Prize biography (2016), Royal Society Fellows profile (2015), and Nature Medicine 10, p.1009 (Oct 2004) profile by Bruce Diamond – consulted for background on Hardy's persona and influence.)

H) FLAGGED "MYTH-RISK" DETAILS

- **Claim:** *"Hardy wasn't convinced the Swedish family had a mutation, so Mullan went ahead and patented it himself."*
  Where it appears: In *Decoding Darkness* (2000), Dr. Mullan is quoted saying he "privately patented" the Swedish APP mutation "because I'd found it, and because [Hardy] hadn't been convinced the Swedish family even had a mutation". This implication – that Hardy showed little faith in the project – has been echoed in some recountings of the episode.
  Why sourcing is weak: The source is a second-hand quote from Mullan, one of the parties involved, presenting his justification. It's not corroborated by independent evidence. In fact, the documented timeline shows Hardy actively facilitated the Swedish study: he invited Lannfelt to London, had his student do linkage on the samples (which *did* show a mutation was likely), and *himself* sent the samples for sequencing in Florida. These actions contradict the notion that Hardy "wasn't convinced" – he was investing lab resources and even entrusting Mullan with the crucial task of sequencing. Mullan's quote seems self-serving, made after the falling-out. No written record from 1991–92 indicates Hardy doubted the family's mutation; on the contrary, Hardy's anger at being excluded

suggests he very much cared about the discovery. Thus, this claim is likely an ex post facto narrative by Mullan, not an objective fact.

Best available evidence: Court findings and team correspondence show Hardy's involvement. Judge Savage's decision notes that sequencing was split in a way that *excluded* Hardy's institution intentionally, implying Hardy was keen enough on the mutation that his colleagues felt the need to cut him (and Imperial College) out. Hardy's own testimony and later commentary express surprise and betrayal – not that he disbelieved the mutation, but that he was left off the paper and patent. In sum, contemporary evidence supports that Hardy was on board with the Swedish project; the "Hardy wasn't convinced" line should be viewed with skepticism. (It's a reminder to rely on documents and multiple accounts rather than a single participant's memory in contentious situations.)

(No other major "myth" anecdotes were identified. Most dramatic elements in Hardy's story – the family inspiration, the amyloid hypothesis gamble, the patent fight – are backed by solid evidence as detailed above.)

LENS RECOMMENDATION

Lens Scorecard:

- **Eureka (Discovery crystallized): 3** – Hardy's identification of the APP mutation is a well-documented "Eureka" moment. We have specific, vivid details: a family-driven quest, a five-year research build-up, and the moment a mutation was found, all with strong sourcing. This discovery scene can be narrated with concrete human and lab details, making for a compelling breakthrough story.

- **Failure (Setback that changed trajectory): 2** – There were setbacks (e.g. the disappointment with the London mutation patent deal and funding, and years of drug trial failures in the field). The patent dispute with Mullan was a significant obstacle, but it frames more as controversy than a personal failure. While it did alter Hardy's path (leaving Imperial), the narrative of Hardy's career is not one of a singular dramatic failure that he had to overcome, aside from persevering through research dead-ends and skeptics.

- **Mentor (Relationship that redirected them): 0** – No pivotal mentor/mentee turning point emerges. Hardy was more of a trailblazer. He had supportive figures (a high-school teacher, PhD advisors) and he himself mentored others (like Alison Goate), but there's no

single mentor relationship that dramatically changed his direction as, say, Max Perutz did for Watson (nothing of that sort here).

- **Controversy (Conflict over credit/ideas): 3** – The story of the Swedish mutation patent is a clear, well-supported controversy with defined parties, stakes, and an outcome. It has drama (a colleague's betrayal, legal showdown) and multiple perspectives (Hardy vs. Mullan) documented. This could certainly drive a narrative lens focusing on conflict in science.

- **Human Detail (Revealing personal insight): 2** – We have some rich personal details (e.g. his motivation to help a family, his outspoken nature, the fact that he's persisted for decades). These can illuminate his character – for instance, Carol Jennings's influence shows his empathy, and the Brexit stance shows his passion for science beyond the bench. However, the narrative is still primarily science-driven. The human elements would support another lens rather than stand alone as the main lens.

- **Last Experiment (Final act or closure): 3** – Hardy's reaction to the 2022 drug success provides a powerful "closure" scene. It's like the epilogue where decades of work pay off. This late-career moment is very scene-worthy (Hardy literally says it's the beginning of the end of Alzheimer's – a dramatic, scene-setting quote) and is strongly sourced. It can serve as a framing device (start or end of the story) to show the arc of his impact.

**Primary recommended lens: ** *EUREKA (Discovery).*
Why it fits: The eureka/discovery lens is the strongest because Hardy's career-defining moment – finding the Alzheimer's gene – is exceptionally well supported and narratively rich. Scene #1 details a clear breakthrough with a before/after impact: before, AD had no known cause; after, a cascade of understanding followed. We have personal texture (a worried family spurring the hunt, Hardy's team working for years) and scientific drama (suddenly a mutation on a gel or sequencing read-out changes everything). This lens allows the profile to highlight how a biochemical discovery was made and why it mattered, which is perfect for an intro biochemistry audience. It's supported by **Scene 1** (the anchor discovery) and can be reinforced by **Scene 4** (showing the long-term importance of that discovery, as the "eureka" eventually led to a therapy). The discovery lens keeps the focus on scientific insight and problem-solving, aligning with the course context.

Best opening scene: Scene #1 – *The First Alzheimer's Mutation Discovered.* It plunges readers into the narrative: a young geneticist in the 1980s confronted with a medical mystery and an eager family, leading to the "lightbulb moment" when the mutation is found. This scene is both human and scientific, engaging readers from the start.

One quote to use early: Quote #1, where Hardy speaks about Carol's family – *"Throughout her family were people who developed Alzheimer's in their 50s…she wanted answers. She's a very determined woman."*. This quote can open the profile to immediately humanize the discovery: it shows why Hardy embarked on this quest and foreshadows the breakthrough (the determination to find answers). It also lets Hardy's voice set the tone of empathy and curiosity.

Cautions: When using the Eureka lens, be careful not to oversimplify the story into "one brilliant flash and that's it." In Hardy's case, the discovery was a team effort and a gradual build-up. Our evidence is solid, but we should avoid mythologizing him as a lone genius. Also, acknowledge that the "eureka" didn't solve everything overnight – there were years of work after identifying the gene (the lens can still incorporate that context). No major disputed details about the discovery itself (the APP mutation finding is well established), but one should not imply Hardy did it alone or that others didn't play a role (to stay accurate).

Backup lens: *LAST EXPERIMENT (Culmination/closure).* (Score: Eureka 3 vs. Last Experiment 3 – essentially tied, with the closure scene being extremely compelling; we choose Eureka as primary due to the initial discovery's centrality, but "Last Experiment" is a close second.)

- **Why it fits:** The narrative could equally start from the end: Hardy in 2022, triumphant as the hypothesis he championed is validated. This lens would frame the profile as a retrospective: begin with Hardy's emotional reaction at the drug's success (Scene #4) and then flash back to how it all began (Scene #1), thus book-ending the story. It highlights the theme of perseverance and long-range vision in science. Scene #4 is one of the most vivid, heartfelt moments we have, with Hardy's own jubilant and reflective quotes – a goldmine for narrative. For students, this lens underscores why the discovery mattered by showing its payoff.

- **Best opening scene:** Scene #4 – *The 2022 "historic" trial result*. For example, the profile could open with: *"In a packed conference hall in 2022, Sir John Hardy fought back tears as data on a new Alzheimer's drug flashed on screen…"* – immediately grabbing attention with the climax and then segueing into "How did we get here? It began 30 years earlier in a London lab…".

- **One quote to use early:** Quote #6, Hardy's exclamation: *"This trial is an important first step… the beginning of the end. …30 years so this has been a long time coming."*. This would set an inspirational tone and frame the story around the fulfillment of a long quest.

- **Cautions:** The "Last Experiment" lens should not make the profile too present-focused at the expense of explaining Hardy's contributions. We'd need to smoothly transition from the 2022 scene to the past. Also, since this success is very recent, avoid giving sole credit to Hardy for the drug – he provided the foundational hypothesis, but others developed the

therapy. Ensuring the narrative doesn't overclaim is important (our sources and Hardy himself attribute success to many). As long as the profile uses the 2022 breakthrough as a narrative hook and then correctly attributes the science, this lens would be highly effective, with only minor caution to keep the spotlight on Hardy's role in enabling that breakthrough rather than the drug invention itself.